



Explore data analyst tools

I'm looking forward to introducing you to some of the tools data analysts use each and every day.

There are tons of options out there.

But the most common ones you'll see analysts use are spreadsheets, query languages and visualization tools.

And this video is going to give you a quick look at how these tools are being used by data analysts every day.

Believe it or not, I was several years into my accounting and finance career before I saw all of these tools working together.

At that point I was very experienced with spreadsheets, and had worked in large data sets with some of the traditional database programs.

I had the foundational skill set to use query languages, and

I had **dabbled** in visualizations, but I had never brought them all together.

Then I got hired here at Google.

And it was so eye-opening to come into a place like

this with an abundance of information everywhere you look.

As an analyst at Google, the true power of these tools became so much clearer to me.

I became more focused on really maximizing everything these tools could do, streamlining my reporting and just making my work simpler.

All of the sudden, I had a lot more time and space to dedicate to identifying new problems to solve and driving decision-making.

Without a doubt, once you've learned the power of these tools, you will be well on your way to becoming the best data analyst you can possibly be.

All right; I hope that story has you even more motivated for this course.

Let's get started with spreadsheets.

Again, there are lots of different spreadsheet solutions, but

two popular options are Microsoft Excel and Google Sheets.

To put it simply, a spreadsheet is a digital worksheet. It stores, organizes, and sorts data.

This is important because the usefulness of your data depends on how well it's structured.

When you put your data into a spreadsheet, you can see patterns, group information and easily find the information you need.

Spreadsheets also have some really useful features called formulas and functions.

A formula is a set of instructions that performs a specific calculation using the data in a spreadsheet.

Formulas can do basic things like add, subtract, multiply and divide, but they don't stop there.

You can also use formulas to find the average of a number set.

Look up a particular value, return the sum of a set of values that meets a particular rule, and so much more.

A function is a preset command that automatically performs a specific process or task using the data in a spreadsheet.

That sounds pretty technical, I know, so let's break it down.

Just think of a function as a simpler, more efficient way of doing something that would normally take a lot of time.

In other words, functions can help make you more efficient.

Those are the spreadsheet basics for now.

Later on, you'll see them in action and start working with spreadsheets yourself.

The next data analysis tool is called query language.

A query language is a computer programming language that allows you to retrieve and manipulate data from a database.

You'll learn something called structured query language, more commonly known as SQL.

SQL is a language that lets data analysts communicate with a database.

A database is a collection of data stored in a computer system.

SQL is the most widely used structured query language for a couple of reasons.

It's easy to understand and works very well with all kinds of databases.

With SQL, data analysts can access the data they need by making a query.

Although query means question, I like to think of it as more of a request.

So you're requesting that the database do something for you.

You can ask it to do a lot of different things such as insert, delete, select or update data.

Okay, that's a top level look at SQL.

In a later video, we'll explore it further and use SQL to do some really cool things with data.

Lastly, let's talk about data visualization.

You've learned that data visualization is the graphical representation of information.

Some examples include graphs, maps, and tables.

Most people process visuals more easily than words alone.

That's why visualizations are so important.

They help data analysts communicate their insights to others, in an effective and compelling way.

When you think about the data analysis process, after data is prepared, processed and analyzed, the insights are visualized so it can be understood and shared.

This makes it easier for stakeholders to draw conclusions, make decisions, and come up with strategies.

Some popular visualization tools are Tableau and Looker.

Data analysts like using Tableau because it helps them create visuals that are very easy to understand.

This means that even non-technical users can get the information they need.

Looker is also popular with data analysts because it gives them an easy way to create visuals based on the results of a query.

With Looker, you can give stakeholders a complete picture of your work by showing them visualization data and the actual data related to it.

All visualization tools have great features that are useful in different situations.

Soon you will learn how to decide which tool to use for a particular job.

And that's everything you need to know about the data life cycle and the data analysis process.

You'll get a chance to test out what you know, so you can feel confident moving forward in this course.

Feel free to take some time to re-familiarize yourself with the concepts and when you're ready, give it your best shot.

If you're ever unsure of an answer, you can always go back and review the videos and readings.

Then you'll be ready to move on to the next set of videos,
where we'll continue exploring the data analytics tools you've already covered.
And you'll get some really fascinating insights into exactly how they work.
Before long, you'll have the knowledge and confidence to start using them yourself.
Stay tuned.